

DFG 3D-Viewer as a Distributed Workflow for Publication, Preservation and Reuse

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Abstract

Recent years have seen the growing importance of 3D modelling in the preservation, documentation, and dissemination of cultural heritage. Digital reconstructions provide access to artefacts, monuments, and sites that are physically inaccessible, fragile, remote, or lost, while also supporting analysis, interpretation, and reuse (Capone & Campi, 2014). Yet, despite the growing number of 3D objects produced by heritage institutions and research projects, standardised processes for their documentation, aggregation, and long-term preservation remain insufficiently developed (Manz et al., 2023). Scholars have repeatedly identified critical gaps in 3D heritage infrastructures, including rights management, uncertainty visualisation, version control, metadata quality, interoperability, searchability, and sustainable preservation (Koller et al., 2010).

The DFG 3D-Viewer project responds to this infrastructural gap by developing a distributed, standards-based workflow for preserving, aggregating, and presenting 3D models as cultural heritage research data. During its first phase (2021–2023), the project established the foundational components required to integrate 3D models into the DFG Viewer environment (see Fig. 1). This included the use of the Metadata Encoding and Transmission Standard (METS) and the Metadata Object Description Schema (MODS) to structure metadata exchange and enable aggregation across repositories (Bajena & Beck, 2024). To test this workflow, a prototype repository was implemented in the WissKI environment and connected to a custom Three.js-based viewer, later developed as AIM 3D. The viewer supports a broad range of 3D formats and enables web-based presentation of heterogeneous 3D assets (Dworak, 2024). Together, these components demonstrated that 3D models can be collected, converted, documented, aggregated, and displayed within a library-compatible infrastructure (Bajena & Kuroczyński, 2025). This testing phase also

included a development of an aggregation instance that harvested 3D content from partners repositories and relevant datasets form Sketchfab (Münster, 2023).

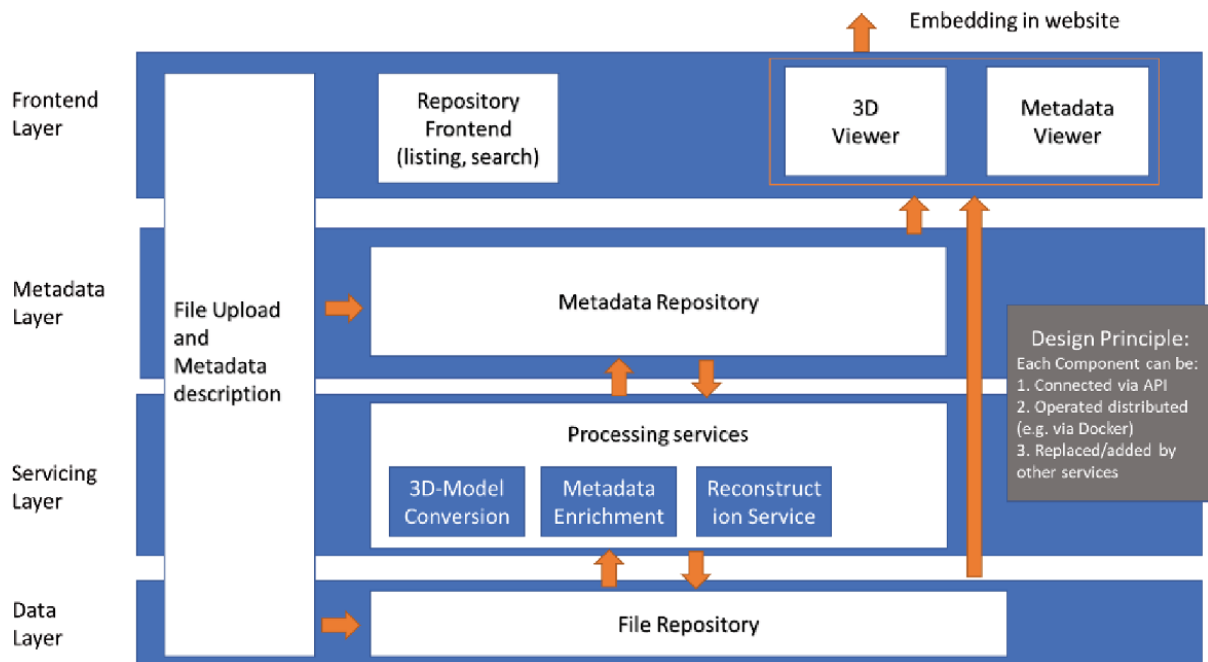


Fig. 1 Software architecture of DFG 3D-Viewer

The second funding phase develops this proof of concept into a more sustainable and institutionally embedded infrastructure (Weigelt, 2025). Its methodological focus combines pragmatic metadata standardisation with semantic modelling. On the pragmatic level, the project refines a METS/MODS application profile for 3D models and tests it with partner repositories. On the semantic level, it advances the WissKI “Recipe for 3D”, a CIDOC CRM-based framework for documenting provenance, technical history, modelling processes, and paradata. At the presentation layer, the project moves beyond a single-viewer solution towards a modular multi-stack ecosystem, allowing different viewers and services to be selected according to data type, use case, and institutional context.

The achieved results show that sustainable preservation of complex 3D assets requires institutionalised workflows rather than isolated project-based tools. Partner implementations confirm that 3D models can be integrated into existing repository systems while supporting discovery, reuse, and long-term archival stability. The project’s outputs – a community-refined METS/MODS profile, the WissKI Recipe for 3D, and a modular viewer infrastructure – therefore function not only as technical components, but also as a practical model for community-owned stewardship of 3D heritage data.

By aligning with emerging initiatives such as IIIF for 3D (Haynes, 2023) and Europeana aggregation protocols, the DFG 3D-Viewer contributes directly to the FAIR 3D Heritage themes of documentation, publication, preservation, and web-based interoperability. It demonstrates how 3D cultural heritage can move from fragmented dissemination towards actionable, standards-based, and sustainable digital preservation.

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